

Month 4: Semantic Re-ranking, LLM Inference, and Prompt Synthesis

This month focused on cross-encoder re-ranking, Groq LLM integration, prompt template design, and citation formatting.

Week 1: Cross-Encoder Re-ranking Integration

- **Progress Summary:** To support multi-hop reasoning, a ColBERT-based cross-encoder re-ranker was integrated into the retriever module. The re-ranker evaluates token-level attention matching between the query and candidate documents, sorting the merged results.

Week 2: Similarity Threshold Filtering

- **Progress Summary:** Dynamic similarity threshold filtering was implemented. Logic was added to discard candidate chunks with scores below the user-defined threshold, optimizing the trade-off between retrieval recall and context noise.

Week 3: LLM Client Integration (Groq)

- **Progress Summary:** The LLM client was integrated using the Groq API, targeting the high-speed Llama-3.3-70b-versatile model. The API connection was configured to stream response tokens back to the server in real-time.

Week 4: Prompt Engineering and Citation Generation

- **Progress Summary:** The prompt templates (prompts.py) were designed. Prompts were engineered to ground responses in the retrieved context and format citations using inline numerical badges. End-to-end integration tests were conducted to verify the pipeline flow.